
SHADOWGRAM

Behavioral Graph Detection of Coordinated AI-Fraud Accounts

"It doesn't just flag an account as suspicious — it shows the evidence behind the suspicion."

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1. Problem Statement

Most fraud- and content-moderation systems reduce a complex judgment to a single opaque number: an account is flagged as some percentage “risky,” with little indication of why. That opacity makes flags hard to trust, hard to contest, and hard for a human reviewer to act on with confidence.

At the same time, coordinated inauthentic behavior — networks of accounts that appear independent but act in concert to spread scams, spam, or influence campaigns — is a well-documented and growing problem. Meta has publicly described investigating coordinated behavior across accounts on its platforms, and fraud-detection vendors such as Experian describe systems that combine behavioral signals (typing patterns, hesitation, copy-paste behavior, navigation) with device and network intelligence to catch fraud that individual account checks would miss.

ShadowGram reframes the detection question. Instead of the difficult, often unanswerable question “Is this account fake?”, it asks a more tractable one:

“Do these apparently independent accounts behave as if they are being operated together?”

2. Proposed Solution

ShadowGram is a behavioral-intelligence prototype that builds a “behavioral fingerprint” for every account in a simulated social-media environment, then looks for hidden coordination between accounts that appear unrelated. It does not stop at a single suspicion score — its centerpiece is an explainability layer that shows exactly which signals a link is based on, turning a black-box score into an inspectable evidence trail.

“ShadowGram does not identify a person. It identifies behavioral relationships between accounts.”

This framing keeps every claim the system makes scientifically and ethically defensible: ShadowGram reports “signals consistent with coordinated operation,” never “this is the same person.”

3. The Five Behavioral Fingerprints

Every account is scored across five independent evidence categories. No single category is enough to raise a flag on its own — coordination is only reported when several agree.

Fingerprint	What it captures
Interaction / Typing	Typing speed and rhythm, hesitation, copy-paste frequency, mouse movement, click intervals, scroll behavior
Navigation	Sequence of pages visited per session (e.g. Home → Search → Profile → Post), session length, pages per session
Temporal	Active hours, posting frequency, interval between actions, and synchronized activity across accounts

Fingerprint	What it captures
Content	Semantic similarity between posts or messages, detected even when the wording differs, using sentence-embedding models
Environment (prototype only)	Browser, OS, screen resolution, timezone, language, and device/network identifiers, collected with consent inside the simulated ShadowGram environment — signals not available on a real platform like Instagram

4. Dashboard & the “Why” Layer

A normal fraud dashboard reports a score. ShadowGram's dashboard is built around explaining that score:

- **Cluster overview** — total accounts analyzed, number of shadow clusters found, and a coordination score for each cluster
- **Behavioral fingerprint card** — a per-account breakdown of typing, navigation, and activity metrics
- **Compare Behavior view** — a side-by-side comparison of two accounts across every fingerprint dimension, ending in an overall similarity percentage
- **“Why are these accounts linked?” panel** — the centerpiece feature — breaks a coordination score into its contributing signals (navigation, timing, typing, posting interval, content), each with a short plain-language explanation. This is what turns a number into evidence
- **Timeline view** — the story of a cluster forming: account creation, first synchronized activity, first near-identical posts, shared suspicious links — ending in “Shadow Cluster Detected.” Built to give a non-technical reviewer a narrative, not just numbers
- **Prevention panel** — recommended platform actions (additional verification, limiting suspicious link distribution, human review, monitoring related new accounts), with a simulated before/after view of contained impact

5. System Architecture

Raw interaction, navigation, and timing signals are converted into a behavioral feature vector per account. An anomaly-detection stage and a similarity engine (enriched with content embeddings) feed a graph engine, which clusters accounts into “shadow clusters” and hands each one to the explanation and prevention layers.

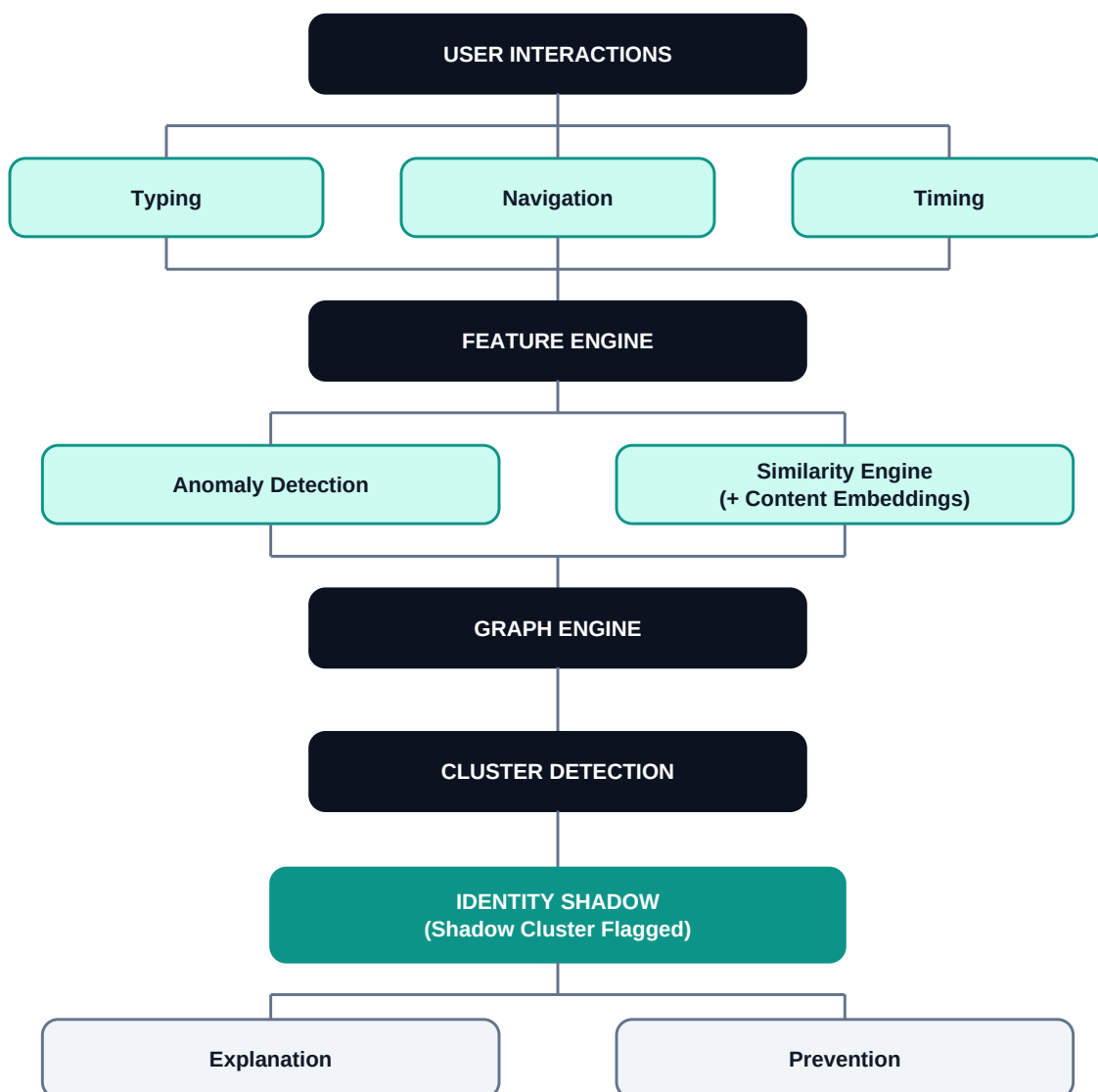


Figure 1. End-to-end ShadowGram detection pipeline.

6. Machine Learning Approach

The first version deliberately avoids deep learning — classical ML is enough to demonstrate the concept convincingly and keeps the system explainable by design:

- **Isolation Forest** — flags unusual individual behavioral patterns

- **Cosine similarity** — compares behavioral vectors between accounts
- **DBSCAN / HDBSCAN** — groups accounts into shadow clusters without pre-setting a cluster count
- **NetworkX** — builds and visualizes the account relationship graph
- **Sentence Transformers** — measures semantic similarity between posted content

Sample behavioral vector

Feature	Account A	Account B
Typing speed (chars/sec)	4.2	4.1
Hesitation (sec)	1.8	1.7
Mouse speed (normalized)	0.73	0.71
Navigation pattern (normalized)	0.91	0.93
Activity start (hr)	20.5	20.2
Activity end (hr)	23.1	23.0
Avg. post interval (min)	47	49
Content similarity	0.88	0.91

Two independently registered accounts whose behavioral vectors are unusually close across every dimension — the pattern ShadowGram is built to surface.

7. Why Multiple Signals, Not One

A system that draws conclusions from a single signal is unreliable: two unrelated people can easily share the same active hours, or a similar typing speed, by coincidence. ShadowGram deliberately requires convergence across typing, navigation, timing, content, and interaction sequence before it raises a cluster — which is what allows it to report a defensible **multi-signal correlation** rather than a coincidence.

8. Recommended Build Plan

Phase 1 — Prototype

Seed the system with a small, controlled dataset: 5 normal accounts, 3 coordinated attacker accounts, and 2 unrelated suspicious accounts. Get the full pipeline working end to end before scaling up:

Login → collect behavior → calculate fingerprint → compare accounts → generate Shadow Cluster → explain why → display graph → simulate prevention.

Phase 2 — Demo scale

Expand the simulated dataset to roughly 100 accounts so the cluster view, graph, and dashboard demonstrate the system at a more realistic scale for the final presentation.

9. Prevention & Impact

When a high-confidence cluster is detected, ShadowGram recommends concrete platform actions rather than just raising an alert:

- Require additional verification from the flagged accounts
- Limit distribution of suspicious links shared by the cluster
- Route the cluster for human review
- Monitor newly created accounts that share the same behavioral signals

Metric	Before intervention	After intervention
Accounts involved	3	3
Suspicious posts	12	distribution restricted
Potential recipients reached	2,431	campaign contained

Simulated impact of catching one coordinated cluster early, as shown on the prevention screen.

10. Ethical & Responsible Framing

- **Careful language** — every output says “signals consistent with coordinated operation,” never a same-person claim
- **Consent-based data** — environment/device fingerprinting only happens inside the simulated, consenting ShadowGram environment — never on scraped real accounts
- **Honest positioning** — ShadowGram is presented as a transparent prototype inspired by a real, publicly documented cybersecurity problem, building on precedents such as Meta's own coordinated-inauthentic-behavior work — not a claim to be first to solve it

11. Conclusion

Most fraud dashboards stop at a score. ShadowGram's differentiator is that every flagged cluster comes with a “Why are these accounts linked?” evidence trail, built from five independent behavioral fingerprints and presented as a clear timeline and graph. That gives a reviewer — or a judge — something they can actually inspect and trust, rather than a black-box number.

References: Meta Newsroom, “Removing Coordinated Inauthentic Behavior” (2020) and “Coordinated Inauthentic Behavior Explained” (2018), about.fb.com.